domain knowledge and suggesting us the right number of categories into which we should cluster the dataset, then to find the right number of clusters for a dataset we will need to run the algorithm multiple times, trying different values for the number of centroids, and comparing the results using some kind of metric (visual comparison is only possible for 2D and 3D datasets). We'll talk about this in section 12.5.

These issues are certainly limiting, although there are domains in which we can assume or even prove that data can be modeled well with spherical clusters; but even in the best case scenario or in intermediate situations where a spherical cluster is still a good approximation, there is another issue that can limit the application of this clustering algorithm: the curse of dimensionality.

12.2.2 The Curse of Dimensionality Strikes Again

We have already bumped into the curse of dimensionality when we described k-d trees in chapter 9: a data structure that works well for low-to-medium-dimensional spaces, but behaves poorly in high-dimensional spaces.

It's not a coincidence that we find this issue again for k-means: this algorithm is a search heuristic that minimizes the Euclidean distance of points to cluster's centroid; in high-dimensions, however:

- The ratio between volume and surface grows exponentially;